Western Ghats Field Validator

Mobile Application for Ground-Truth Validation of Satellite-Derived Land Cover

> **Key Capabilities:** Offline-first architecture | CoRE Stack API integration (8 endpoints) | Forest vs plantation classification | GPS-based observation capture | JSON/CSV export

1. Overview and Interface

The Western Ghats Field Validator is a mobile-first progressive web application that enables field researchers to validate satellite-derived LULC classifications through systematic ground-truth observations. It integrates CoRE Stack watershed APIs, Google Dynamic World, and custom forest typology datasets.

Interface Features: Interactive map with street/satellite base layers | GPS location tracking | Layer toggle with active count | Bottom navigation (Map, Layers, Capture, Guide, Log)

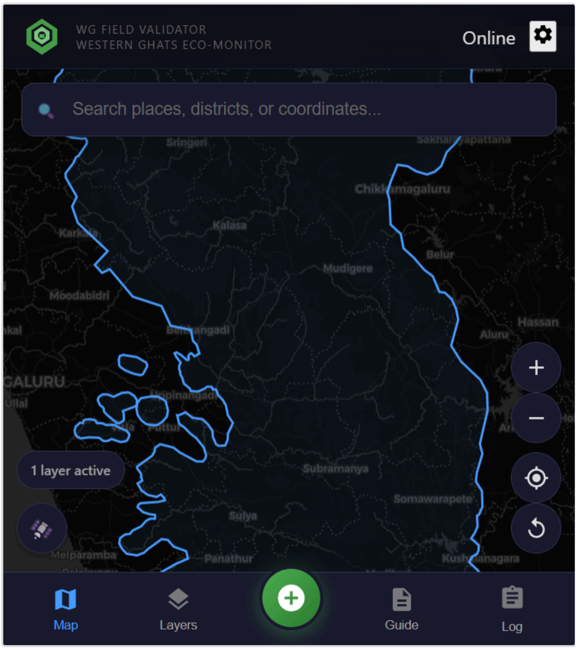

Figure 1: Main interface

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2. Layer Management

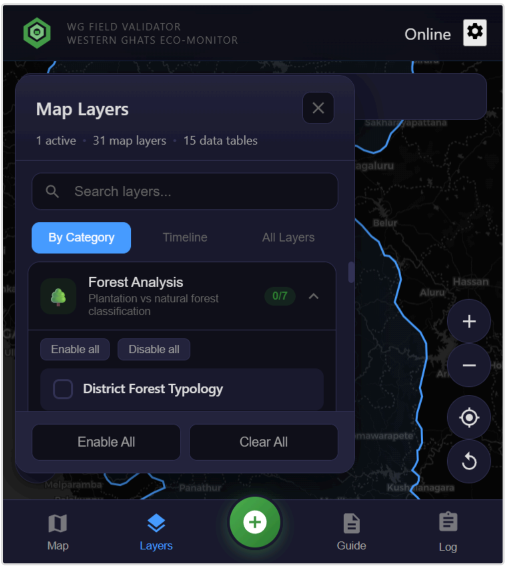

Figure 2: Layer categories

The application organizes **31 map layers** and **15 data tables**:

Category	Description	Count
Forest Analysis	Plantation vs natural forest classification	7
Land Cover Maps	Historical LULC from GLC-FCS30D (1985-2022)	16
Urban Expansion	Built-up area change detection	14
Dynamic World	Google Earth Engine LULC classification	2
Boundaries	Administrative boundaries (District, WG)	2
Watershed Data	Water balance, cropping intensity (CoRE Stack)	4

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3. Forest vs Plantation Classification

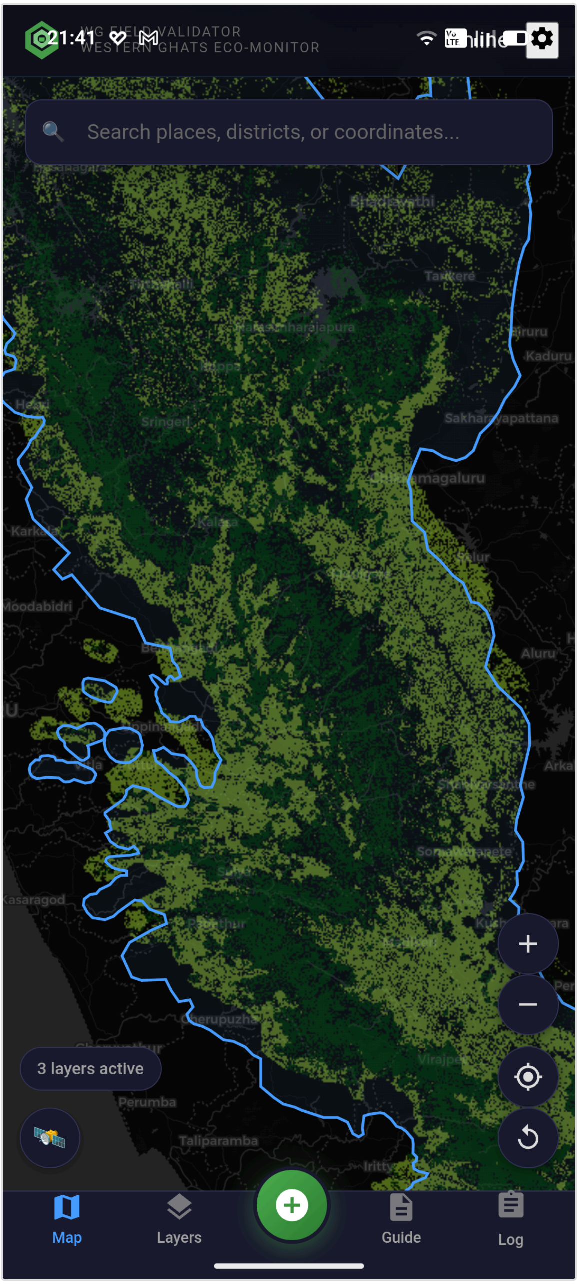


Figure 3: Natural forest (green) vs plantations (purple)

Data Sources

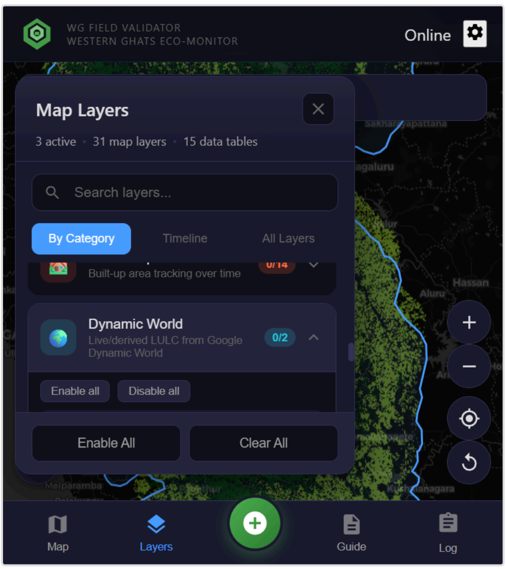
The forest typology layer integrates two pre-existing datasets served locally within the application:

- **Google Dynamic World:** Near real-time LULC classification providing tree cover extent
- **Natural Forest 2020:** AI-based classification separating natural forests from tree plantations, developed by Google Research for deforestation-free supply chain verification (source)

The Natural Forest 2020 dataset uses machine learning trained on global forest inventory data to distinguish between natural forests (multi-species, uneven age) and managed plantations (monoculture, even-aged stands) at 10m resolution.

This distinction is critical: natural forests support significantly higher biodiversity than plantations, despite both appearing as "forest" in conventional satellite classifications.

4. Dynamic World and CoRE Stack Integration



Dynamic World LULC

Google's near real-time land cover with 9 classes: Water, Trees, Grass, Flooded Vegetation, Crops, Shrub/Scrub, Built-up, Bare Ground, Snow/Ice. Available as Live GEE (real-time) or Regional composite (2018-2025).

CoRE Stack API Endpoints

Endpoint	Data Provided
`get_admin_details_by_latlon`	State, District, Tehsil lookup
`get_mwsid_by_latlon`	Micro-Watershed ID
`get_generated_layer_urls`	GIS layers for tehsil
`get_tehsil_data`	Tehsil-level statistics
`get_mws_data`	ET, Runoff, Precipitation time series
`get_mws_kyl_indicators`	Watershed indicators
`get_waterbodies_data_by_admin`	Waterbody inventory
`get_mws_report`	MWS report URL

Figure 4: Dynamic World options

All endpoints tested in `docs/notebooks/corestack_api_tested.ipynb`

5. Field Observation Workflow

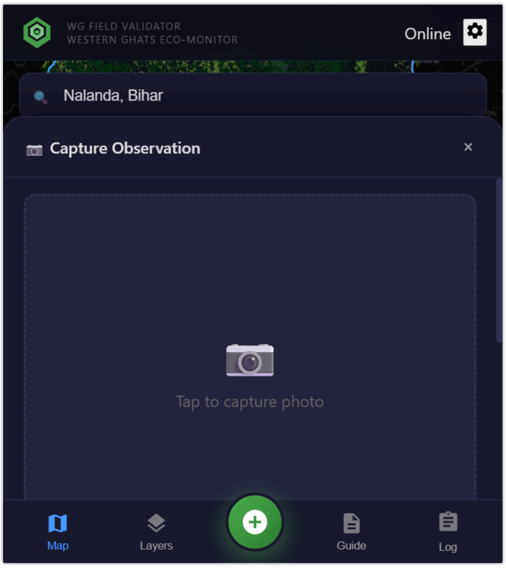


Figure 5: Capture interface

Capture Process

1. **Photo capture:** Camera or gallery selection
2. **GPS coordinates:** Automatic with accuracy indicator
3. **Layer values:** Active layer data at location displayed
4. **Validation status:** Match (agrees) | Mismatch (differs) | Unclear
5. **Field notes:** Optional text documentation

Observations are stored locally in IndexedDB and sync when connectivity is restored.

6. Field Log and Data Export

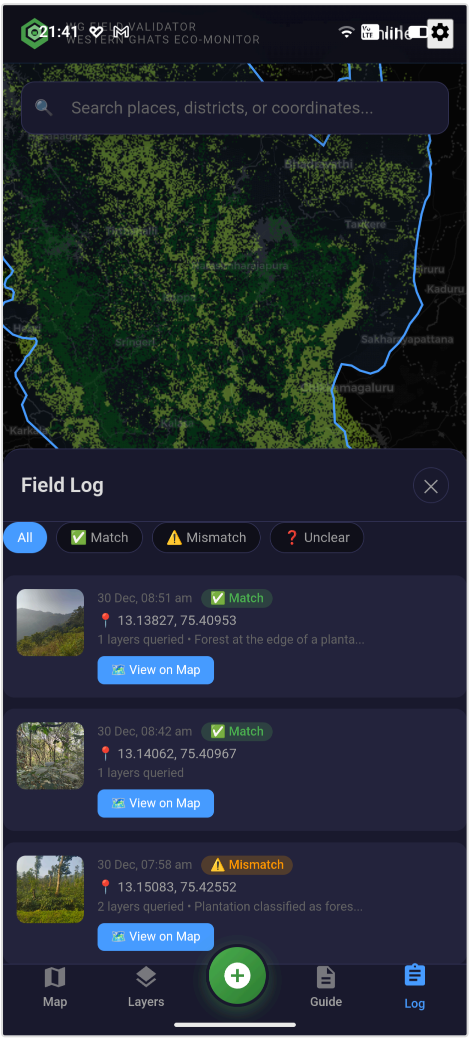


Figure 6: Field log with filtering

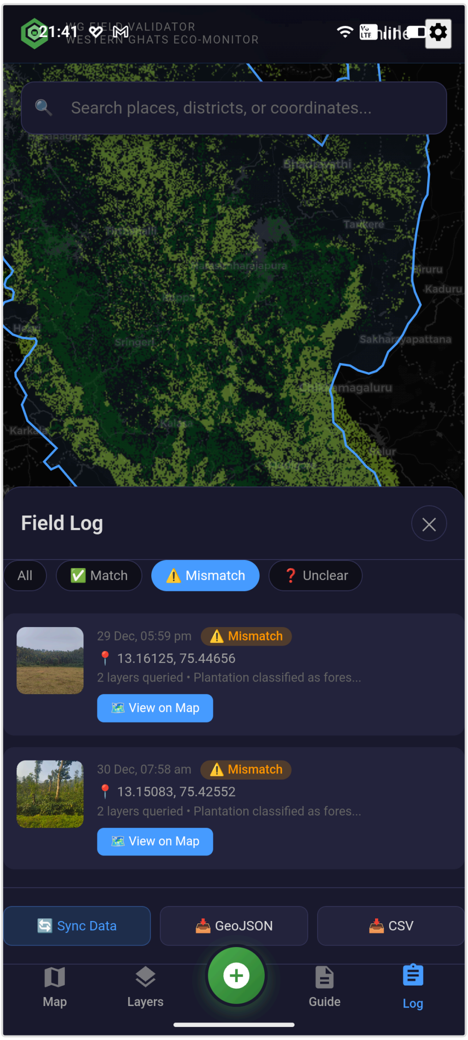


Figure 7: Export options

Log Features

- Chronological list with thumbnails
- Filter by status (All/Match/Mismatch/Unclear)
- View details and map location

Export Formats

- **JSON:** Full metadata for analysis
- **CSV:** Spreadsheet/GIS compatible

7. Additional Features and Technical Architecture

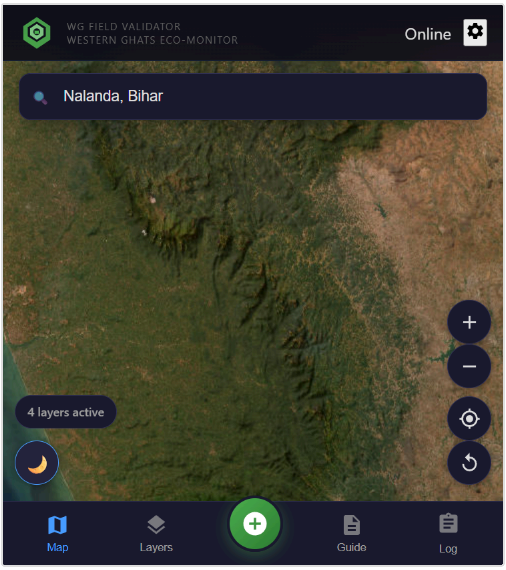


Figure 8: Satellite base map

Base Maps

- OpenStreetMap: Navigation
- ESRI Imagery: Verification
- Dark mode: Low-light

Field Guide

- Getting started / GPS setup
- Layer management
- Offline operation

Technical Stack

Frontend	React 18 + TypeScript + Vite
Mapping	MapLibre GL JS
Mobile	Capacitor (Android)
Storage	IndexedDB
APIs	CoRE Stack, Nominatim

8. Bidirectional Value: Field Data and Geospatial Models

Geospatial Data Enriches Field Work

Field investigators can leverage satellite-derived datasets and models to enhance their data collection:

- Pre-visit planning using LULC classifications to identify target areas
- Real-time context from CoRE Stack watershed indicators at any location
- Historical change detection layers to prioritize validation sites
- Multi-layer comparison to cross-reference classifications during fieldwork

Field Data Refines Geo-Models

Ground-truth observations collected through this app can programmatically improve model accuracy:

- Structured JSON/CSV exports compatible with ML training pipelines
- GPS-tagged photos provide labeled samples for classification refinement
- Match/Mismatch validation flags identify systematic model errors
- Offline collection enables data gathering in remote, under-sampled regions

Resources

Sample data: `field-data/recovered-observations/` | API notebook: `docs/notebooks/corestack_api_tested.ipynb` | License: MIT